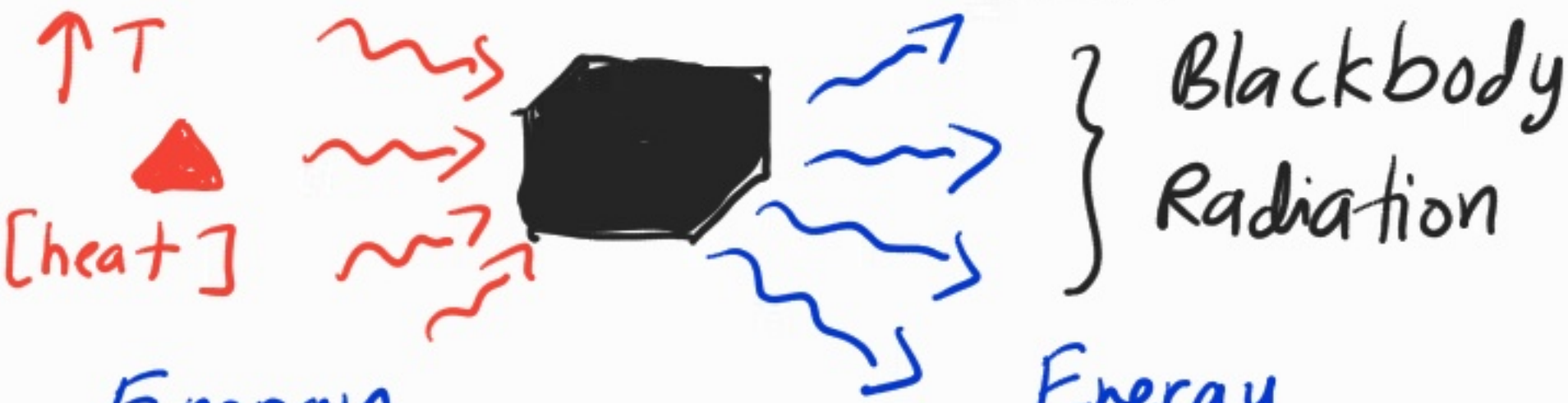
Emission & Absorption of Light

Black Body → objects that absorb & re-emit large amounts of energy that is incident upon it.

Example → wearing a black shirt on a hot day.

PERFECT BLACKBODY



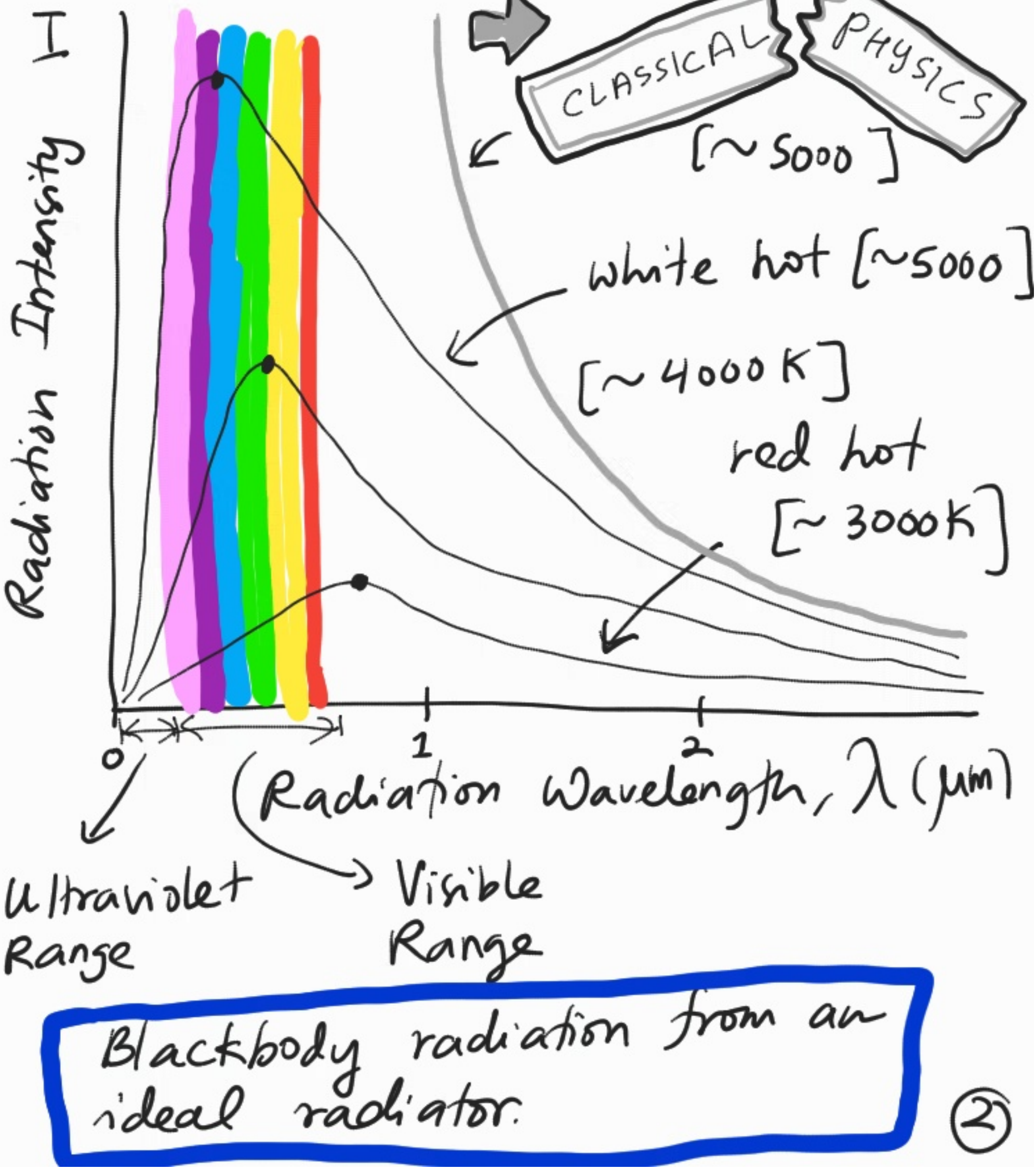
Energy Absorbed (E_1)

Energy Emitted (E_2)

$$E_1 = E_2$$

Emission & Absorption of Light

⇒ Electromagnetic Intensity Graph.



Blackbody radiation from an ideal radiator.

Emission & Absorption of Light

- Rayleigh-Jeans law:

$$I(T) = \frac{2ck_BT}{\lambda^4}$$

$c \sim$ speed of light (m/s)

$k_B \sim$ Boltzmann's constant $(1.381 \times 10^{-23} \, J \, K^{-1})$

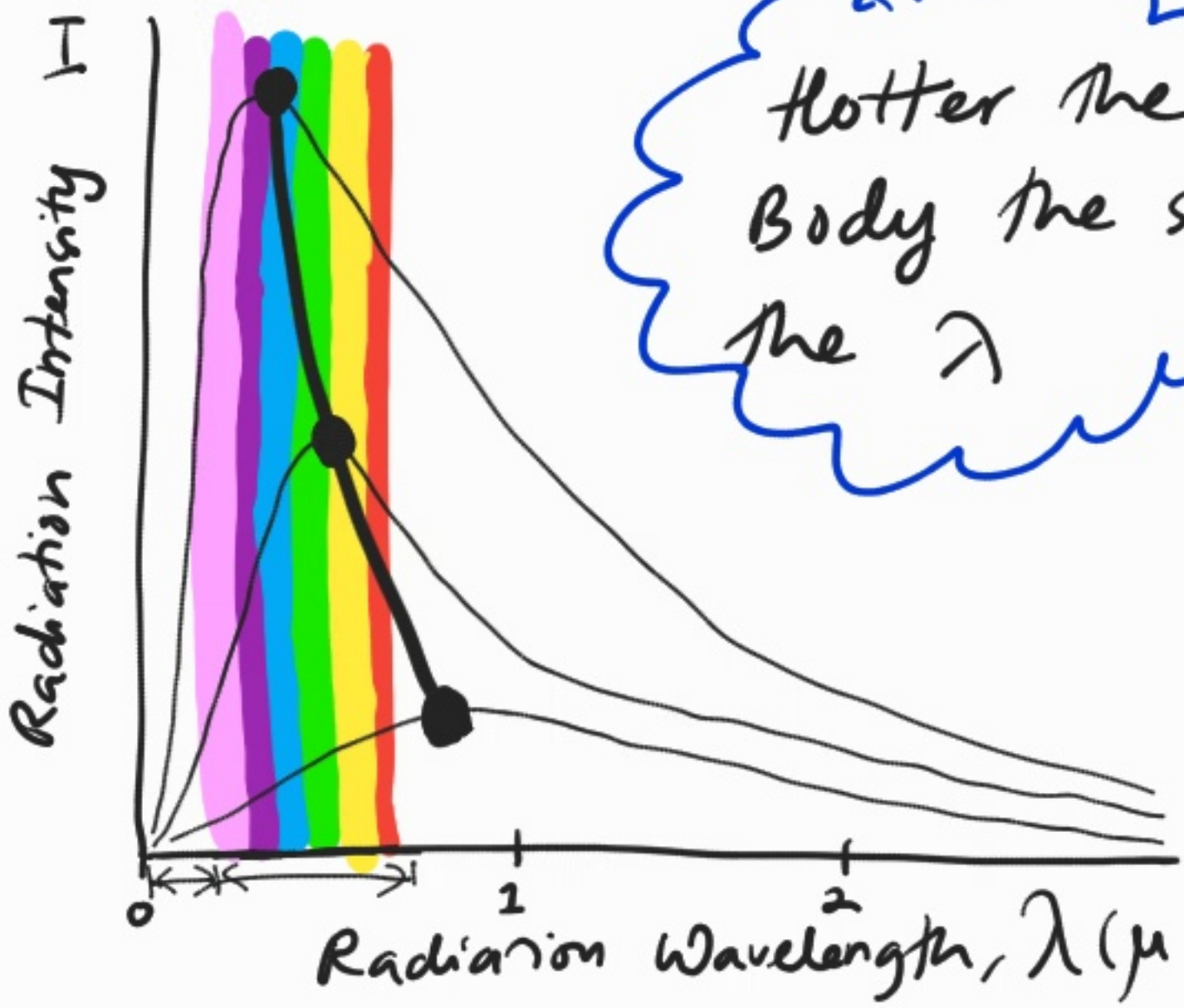
$\lambda \sim$ wavelength (m)

$T \sim$ temperature in K.

Ultraviolet Catastrophe

Emission & Absorption of Light

- 2 Laws for the experimental findings of blackbody radiation.



Hotter the Body the shorter the λ

① Wien's Displacement Law:

$$\lambda_{max} T = 2.898 \times 10^{-3}\ m \cdot K$$

Emission & Absorption of Light

(2) Stefans Law:

- Total power of radiation by a blackbody emitted across the entire spectrum of λ at a specific temperature.

$$P(T) = \sigma A T^4$$

A ~ surface of the blackbody (m^2)

T ~ temperature in K

σ ~ Stefan-Boltzmann constant. [$W/(m^2 \cdot K^4)$]

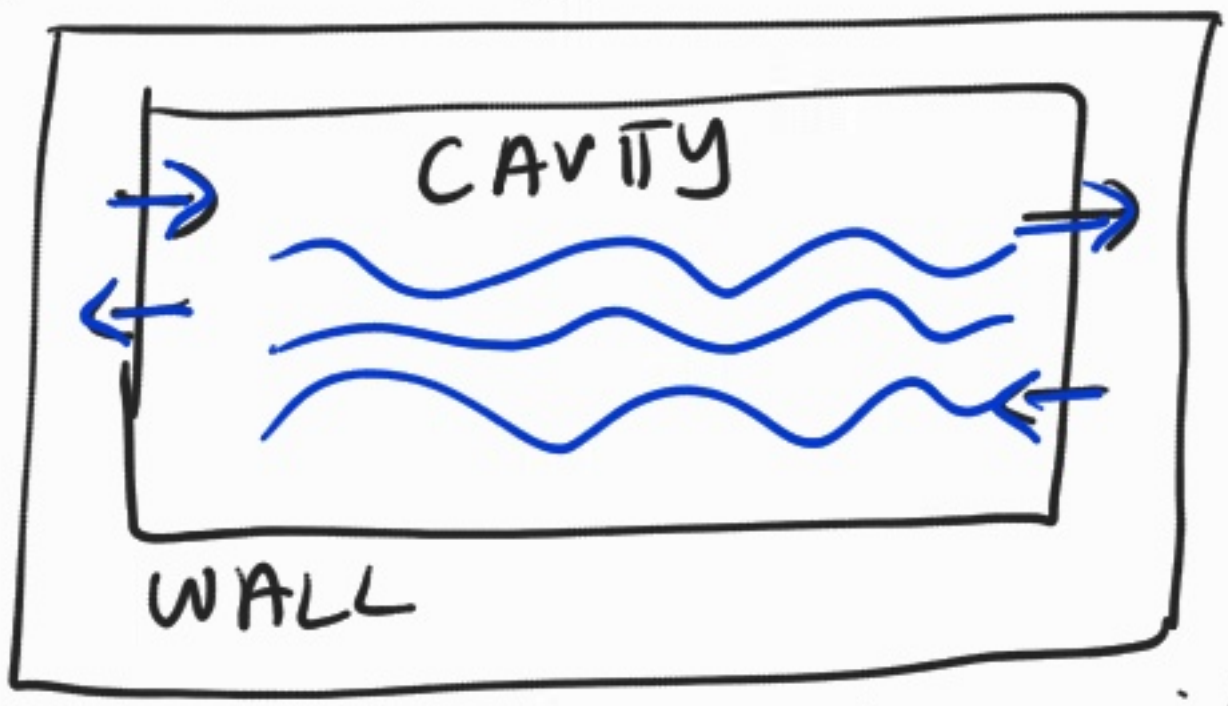
$$\sigma = 5.67 \times 10^{-8}\ W/(m^2 \cdot K^4)$$

Emission & Absorption of Light

- classical approach, radiation is treated as waves.
- Unfortunately divergence between theoretical & experimental findings result in UV catastrophe for blackbody radiation phenomena.

CLASSICAL PICTURE.

EM waves trapped in a cavity.

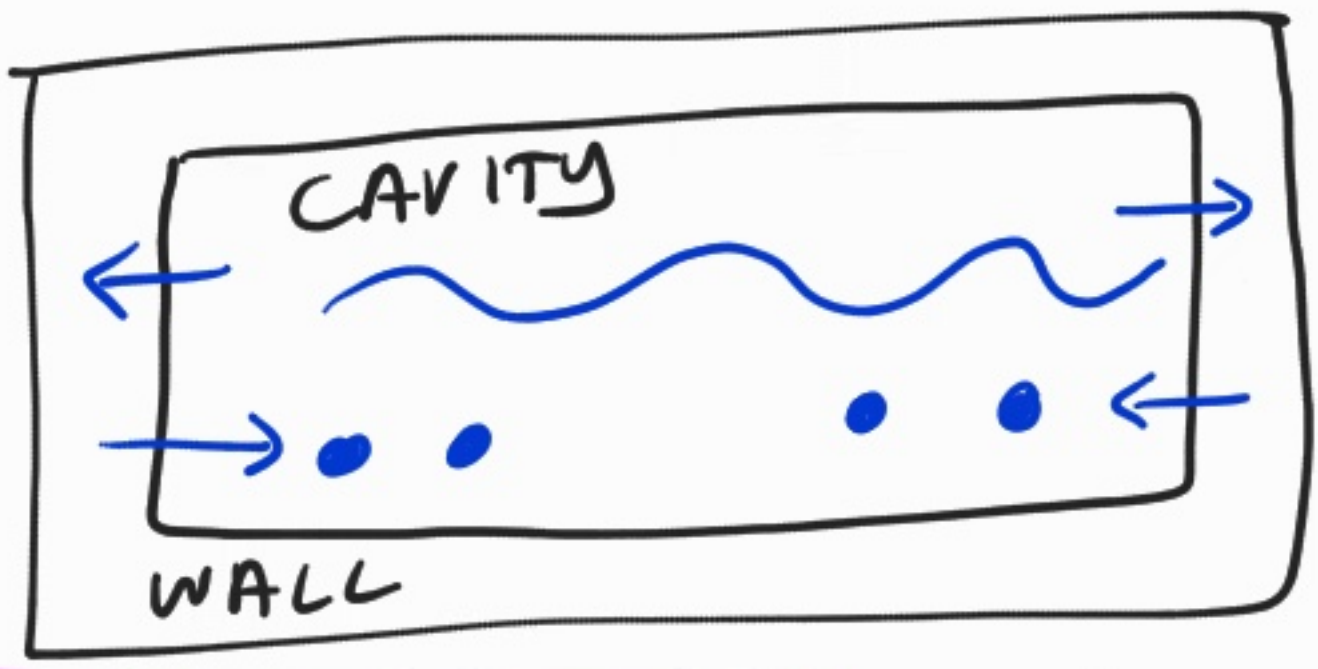


☞ EM waves are in equilibrium with wall.

Emission & Absorption of Light

Planck's Hypothesis.

⇒ Cavity radiation originates from atomic oscillations inside the cavity walls that produce discreet values of energy.



Discreet Energy values ⇒ Quanta [quantized energy]

Emission & Absorption of Light

- Planck assumed ~~that~~ the energy of an oscillator (E_n) can only have discreet (quantized) values:

$$E_n = nhf \quad \text{where } n = 1, 2, 3, \cdots$$

h - Planck's constant

$(4.136 \times 10^{-15}\ eV \cdot s)$

n - quantum number

→ n describes the quantum state of a Planck oscillator.

Emission & Absorption of Light

- Planck's Quantum Hypothesis of energy quanta :

$$\boxed{\Delta E = hf}$$

↳ energy jump between 2 consecutive quantum states.

$$\{hf, 2hf, 3hf, \dots, nhf\}$$

- Planck's Blackbody Radiation

$$\underbrace{I(\lambda, T)}_{\text{power intensity}} = \frac{2\pi hc^2}{\lambda^5} \cdot \frac{1}{\exp\left[\frac{hc}{\lambda k_B T}\right] - 1}$$

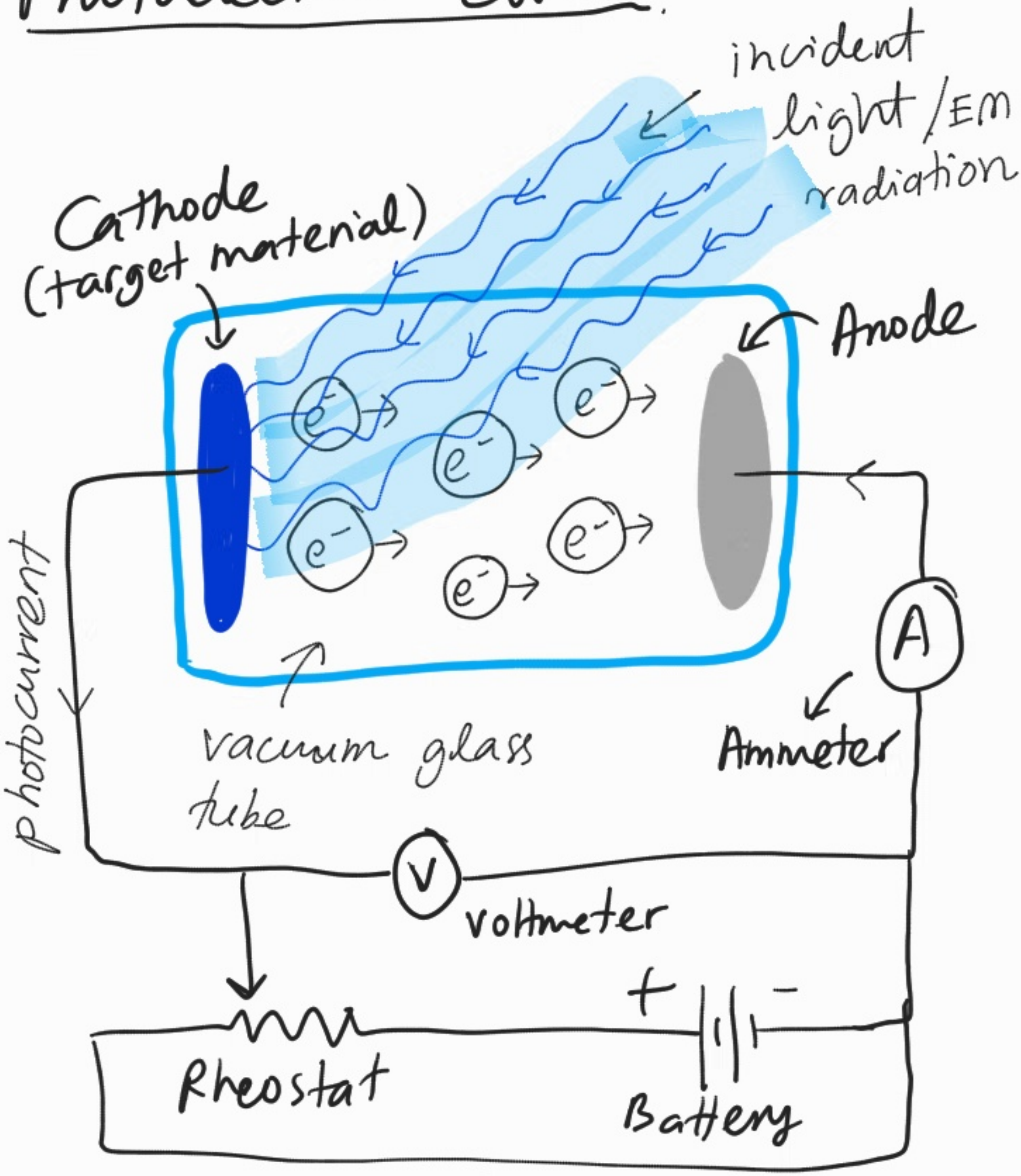
Note: Wien's Law & Stefan's Law can be derived from Planck's BB radiation.

Photoelectric Effect

- A metal surface is exposed to a monochromatic EM wave of sufficiently short λ.
- Then the incident radiation is absorbed by the metal and the exposed surface emits electrons.

This process is called the photoelectric effect & the emitted electrons are called photoelectrons.

Photoelectric Effect



- Reverse potential difference
  → Stopping potential. ✓

Photoelectric Effect

3 key characteristics

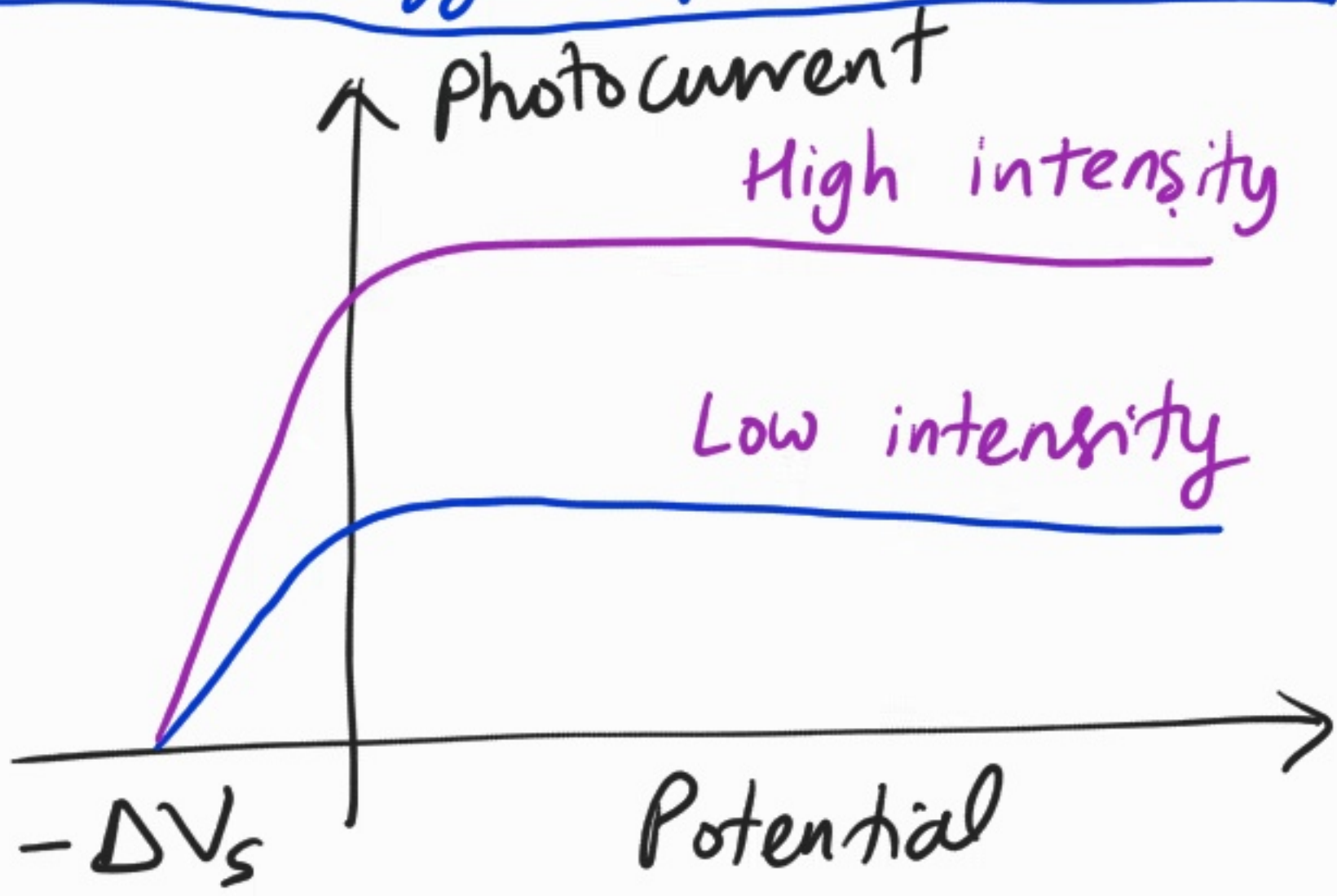
① Absense of Lag time

⇒ instantaneous emission of e^-.

⇒ classical physics predicts energy build-up → lag time.

Photoelectric Effect

(2) Intensity of Incident Radiation ∄ Kinetic Energy of photoelectrons.



[stopping potential]

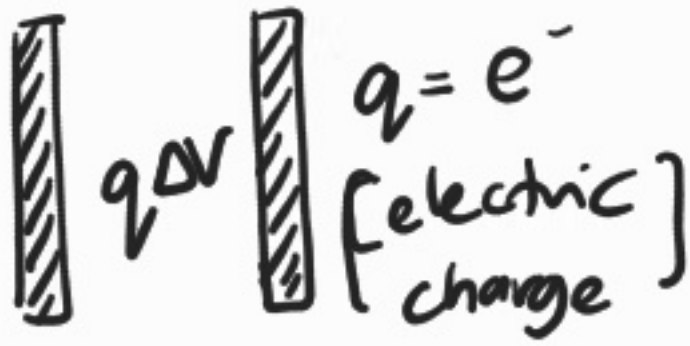
↳ always at one value for any intensity.

Cont

Photoelectric Effect.

⇒ photoelectron leaving the surface has kinetic energy, K.

⇒ it gained energy from the EM wave.



Energy Balance:

$$\Delta K = q\,\Delta V$$

$$\Delta K - e\,\Delta V = 0$$

⇒ Apply stopping potential.

$$\Delta K - e(-\Delta V_s) = 0$$

$$(0 - K_i) + e\,\Delta V_s = 0 \rightarrow K_i = e\,\Delta V_s$$

∴ largest kinetic E in the presence of the stopping potential:

$$K_{max} = e\,\Delta V_s$$

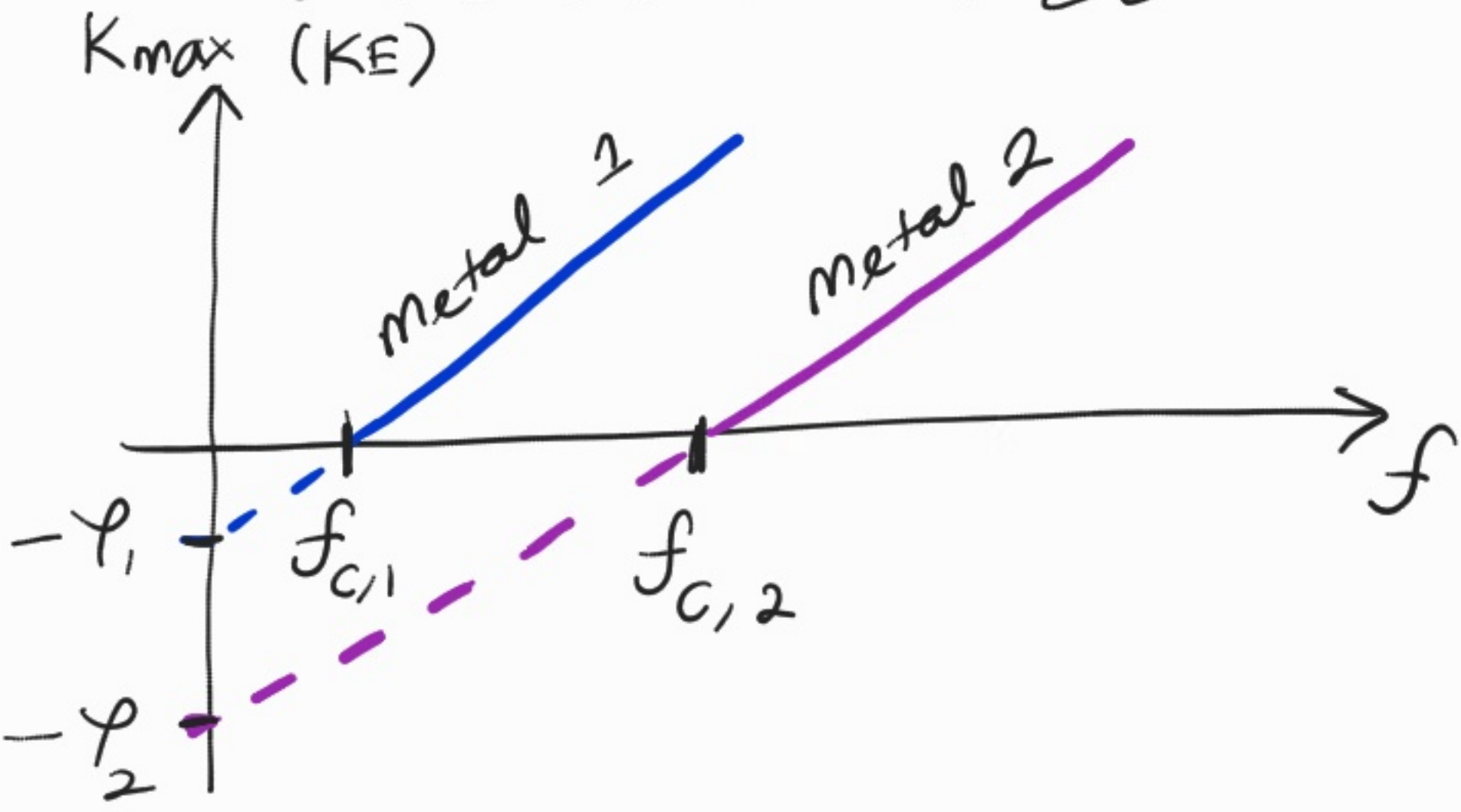
- Classical Theory:

Intensity EM radiation $\propto$ Kinetic Energy of photoelectrons.

Photoelectric Effect

③ Presence of Cut-off Frequency.

- The photoelectric Effect Experiment show that there exists a minimum frequency of the incident radiation – where below this minimum frequency, NO PHOTOCURRENT IS GENERATED.



Con't

Photoelectric Effect.

Contradictions with classical Physics:

⇒ Kinetic energy of photoelectrons is independent of the frequency of the incident radiation.

⇒ There should be no cut-off frequency.

⇒ Electrons should receive energy in a continuous fashion.

⇒ Kinetic energy of photoelectrons should depend exclusively on the intensity of incident light.